

Certifying Reconstruction from Finite Data

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Abstract

We study finite-sample reconstruction in measure-preserving systems. Given a time series $x_1, \dots, x_n$ from a system $(X, \mathcal{B}, \mu, T)$ and a single observable h , we ask whether reconstruction — the condition that the delay algebra $\sigma(h, h \circ T, \dots)$ generates $\mathcal{B}$ modulo μ — is detectable from data alone, at what rate, and with what certificates.

We prove three theorems. The *Algebra Theorem* shows that the empirical σ -algebra approximation error $\hat{\delta}(L, n)$ concentrates around the true error $\delta(L)$, and that the elbow stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d -dimensional state space, without oracle knowledge of the mixing rate, the lag, or the smoothness index. The *Dynamics Theorem* shows that under uniform separation, the delay map $\Phi_h^{(L)}$ is bi-Lipschitz and estimated delay vectors certify point separation at rate $n^{-\beta/(2\beta+d)}$. The *Conjunction Theorem* shows that for deterministic T , algebra separation and metric separation are the same event — not merely correlated, but identical — and that both witnesses certify this from data under reconstruction and uniform separation.

A third computable witness emerges from the fibre structure of the delay map. Under the fibre mixing condition — that ν_L -almost every large fibre is balanced — the collision entropy $H_2(\nu_L) := -\log(\mu \otimes \mu)(R_L)$ diverges if and only if $\delta(L) \rightarrow 0$. The key is a conditional variance identity expressing $\delta(L)$ as an integral of within-fibre variance over unseparated trajectory pairs; the identity reverses under fibre mixing but fails when large fibres are monochromatic — entirely inside or outside the hardest-to-approximate event. All three witnesses — algebraic, geometric, and information-theoretic — measure the same failure of separation across delay fibres.

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1 Introduction

The reconstruction theorem states that the observable algebra $\mathcal{O}_h = \sigma\{h \circ T^n : n \in \mathbb{Z}\}$ generates $\mathcal{B}$ modulo μ if and only if the delay map Φ_h is a measure-theoretic embedding — an exact asymptotic condition that says nothing about rates, certificates, or what can be verified from a finite time series. Given a time series $x_1, \dots, x_n$ from $(X, \mathcal{B}, \mu, T)$ and a single observable h , we ask: can one determine from data alone whether reconstruction holds, at what rate, and with what certificates? We answer with three theorems: the Algebra Theorem, the Dynamics Theorem, and the Conjunction Theorem. Both results [5, 4] are taken as given; this paper asks what finite data can certify.

The move from reconstruction to certification is a move from measure theory to statistics, and it introduces demands that the abstract reconstruction theorem does not. The witnesses — the empirical approximation error $\hat{\delta}(L, n)$ and the kernel separation — are well-defined for any measure-preserving system and any $h \in L^\infty$. What requires hypotheses is *concentration*: to prove that $\hat{\delta}(L, n)$ tracks $\delta(L)$ with quantitative rates, one needs the orbit to mix fast enough that empirical averages concentrate, and the geometry of the delay map to be regular

enough that bias is controlled. We work throughout in the *exponentially mixing regime on compact smooth manifolds* — the setting where minimax-optimal rates are achievable and the witnesses can be shown to certify reconstruction with explicit, data-computable guarantees. The polynomial-mixing case and singular state spaces (e.g. fractal attractors) remain the principal open extensions; the discussion section records precisely where each proof breaks and what would be needed to close the gap.

The Algebra Theorem (Section 5). Define the σ -algebra approximation error

$$\delta(L) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \Delta E),$$

which measures how much of $\mathcal{B}$ is not yet captured by the finite-lag algebra $\mathcal{O}_h^{(L)} = \sigma(h, h \circ T, \dots, h \circ T^L)$. This quantity is estimable from data as $\hat{\delta}(L, n)$ (defined in Section 2). The elbow stopping rule $\hat{L}^*$ fires when $\hat{\delta}(L, n)$ stops decreasing. Under exponential mixing with resolvability constant $C_\lambda > 4$, $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ for Hölder(s) targets on a d -dimensional state space, without any oracle knowledge of the mixing rate, the lag $L^*(n)$, or the smoothness index s . This is the main finite-sample rate result.

The Dynamics Theorem (Section 6). The delay map $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$ encodes the trajectory of x under the observable. When $T \in C^r$ and the uniform separation condition (US) holds (infimum of the smallest singular value of $D\Phi_h^{(L)}$ over X is bounded away from zero), $\Phi_h^{(L)}$ is bi-Lipschitz. The empirical delay vectors then constitute a second witness: they certify that points are being separated by the dynamics. The Dynamics Theorem shows that this kernel witness converges to the true separation at rate $n^{-\beta/(2\beta+d)}$ and names the places where it can fail.

The Conjunction Theorem (Section 7). The two witnesses measure the same event. For deterministic T , the TV separation $d_L(x, x') = 1$ if and only if x and x' are separated by $\mathcal{O}_h^{(L)}$ — not merely correlated, but identical. This algebraic identity ($\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)}$) is the Markov bridge. Under reconstruction and (US), both witnesses fire correctly and certify the same object. The theorem also names two failure modes: reconstruction failure makes the algebra witness fail; violation of (US) makes the kernel witness fail even when reconstruction holds.

The σ -algebra approximation error $\delta(L)$ is observable without knowing the mixing rate, the dimension d , or the smoothness s . The elbow rule for $\hat{\delta}(L, n)$ operates upstream of the rate: it asks whether the algebra has captured $\mathcal{B}$, not at what rate the approximation is converging. In the exponentially mixing regime, for deterministic T satisfying reconstruction and (US), all three witnesses certify the same object with explicit, data-computable guarantees.

2 Setup and Notation

2.1 The measure-preserving system

Let $(X, \mathcal{B}, \mu)$ be a probability space where X is a compact smooth d -dimensional Riemannian manifold and μ is a Borel probability measure with smooth, everywhere-positive density $\rho : X \rightarrow \mathbb{R}_{>0}$ satisfying

$$\rho_{\min} := \operatorname{ess\,inf}_X \rho > 0.$$

Let $T : X \rightarrow X$ be a μ -preserving measurable map. We make no injectivity or invertibility assumption on T beyond what is stated in each hypothesis.

2.2 The observable and the delay map

Fix an observable $h \in L^\infty(\mu)$ with $\|h\|_{L^\infty} =: a < \infty$. For each lag depth $L \geq 0$ define the *delay map*

$$\Phi_h^{(L)} : X \rightarrow \mathbb{R}^{L+1}, \quad \Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x)).$$

The *observable σ -algebra at lag L* is

$$\mathcal{O}_h^{(L)} := \sigma(\Phi_h^{(L)}) = \sigma\{h \circ T^k : 0 \leq k \leq L\}.$$

The *delay-polynomial class* of degree $D \geq 1$ is

$$\mathcal{A}_h^{(L)} := \{p \circ \Phi_h^{(L)} : p \text{ is a polynomial of degree } \leq D \text{ on } \mathbb{R}^{L+1}\} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu).$$

Since $\Phi_h^{(L)}(X) \subseteq [-a, a]^{L+1}$, all elements of $\mathcal{A}_h^{(L)}$ are bounded.

2.3 The approximation error

The central quantity is the σ -algebra approximation error

$$\delta(L) := \delta(\mathcal{O}_h^{(L)}) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E),$$

which measures how much of $\mathcal{B}$ remains outside $\mathcal{O}_h^{(L)}$. It is non-increasing in L , vanishes if and only if reconstruction holds, and under (G1)–(G2) on a smooth manifold reaches zero by lag $L_0 \leq d - 1$ [4].

2.4 Fibre structure and collision probability

The delay map $\Phi_h^{(L)}$ partitions X into *fibres*: for each $z \in \mathbb{R}^{L+1}$, define

$$F_z := (\Phi_h^{(L)})^{-1}(z), \quad p_z := \mu(F_z), \quad \mu_z := \mu(\cdot \mid F_z).$$

Call z an ε -*large fibre* if $p_z \geq \varepsilon$. The push-forward $\nu_L := (\Phi_h^{(L)})_* \mu$ is the distribution of delay vectors.

The set of unseparated pairs $R_L := \{(x, x') : \Phi_h^{(L)}(x) = \Phi_h^{(L)}(x')\}$ decomposes as $R_L = \bigsqcup_z F_z \times F_z$, giving

$$(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z). \tag{1}$$

This quantity is the *collision probability* of the delay-vector distribution: it is the probability that two independent draws from μ produce the same delay vector.

2.5 The data model

We observe a time series $x_1, x_2, \dots, x_n$ which is an orbit segment of T : $x_i = T^{i-1}x_1$ for some initial condition $x_1 \in X$ drawn from μ . The empirical measure is $\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$. We do not assume ergodicity or density of the orbit; each result states the mixing hypothesis it uses.

2.6 Structural hypotheses

The following hypotheses are referenced throughout. They are not assumed globally; each theorem states exactly which subset it uses.

(G1) *No short periodic orbits*: T has no periodic orbit of period $\leq L$ in $\text{supp}(\mu)$. This ensures the orbit points $x, Tx, \dots, T^L x$ are distinct μ -a.e., which is used in the Sard–Smale argument for full rank of $\Phi_h^{(L)}$.

(G2) *Non-degenerate observable*: $h \in C^2(X)$ with $dh \neq 0$ μ -a.e. This places h in the Sard–Smale regular-value residual set and ensures $\Phi_h^{(L)}$ has rank $\min(L+1, d)$ generically.

(US) *Uniform separation*: $\inf_{x \in X} \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, where $\sigma_{\min}$ denotes the smallest singular value. This is a uniform lower Lipschitz bound on $\Phi_h^{(L)}$, strictly stronger than (G1)–(G2) which give full rank only μ -a.e. Condition (US) is used only in the Dynamics and Conjunction theorems (Sections 6 and 7).

2.7 The empirical witness

The empirical analogue of $\delta(L)$ is

$$\hat{\delta}(L, n) := \sup_{p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1} \|p - \hat{\mathbb{E}}[p | \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2,$$

where $\mathcal{P}_D^{(L)}$ is the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$ (normalised in $L^2(\mu)$), and $\hat{\mathbb{E}}[p | \Phi_h^{(L)}]$ is the Nadaraya–Watson kernel regression estimator (Definition 4.5).

The main result of Section 4 is that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$ at rate $\varepsilon_n = C n^{-2s/(2s+d)}$ under exponential mixing, provided $d > 2s$.

3 Bias Bound and Conditional Variance

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub- σ -algebra. Define the *σ -algebra approximation error*

$$\delta(\mathfrak{m}) := \sup_{S \in \mathcal{B}} \inf_{E \in \mathfrak{m}} \mu(S \triangle E). \quad (2)$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ -null sets.

Lemma 3.1 (Conditional variance identity). *For any $S \in \mathcal{B}$:*

$$\mathbb{E}[\text{Var}(\mathbb{1}_S | \mathcal{O}_h^{(L)})] = \frac{1}{2} \int_{R_L} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 d(\mu \otimes \mu)(x, x').$$

Proof. By disintegration, $(\mu \otimes \mu)|_{R_L} = \int \mu_z \otimes \mu_z d\nu_L(z)$. On each fibre F_z :

$$\frac{1}{2} \int_{F_z \times F_z} |\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 d(\mu_z \otimes \mu_z) = \mu_z(S) \mu_z(S^c).$$

Integrating with weight p_z : $\frac{1}{2} \int_{R_L} |\mathbb{1}_S - \mathbb{1}_{S'}|^2 d(\mu \otimes \mu) = \int p_z \mu_z(S) \mu_z(S^c) d\nu_L(z)$. Since $\text{Var}(\mathbb{1}_S | \mathcal{O}_h^{(L)})(x) = \mu_z(S) \mu_z(S^c)$ on F_z , both sides equal $\int \mu_z(S) \mu_z(S^c) d\nu_L(z)$. $\square$

Corollary 3.2 (Easy direction). $\delta(L) \leq \frac{1}{2} (\mu \otimes \mu)(R_L)$.

Proof. In Lemma 3.1, bound $|\mathbb{1}_S(x) - \mathbb{1}_S(x')|^2 \leq 1$ and take the supremum over S using $\delta(L) = \sup_S \mathbb{E}[\text{Var}(\mathbb{1}_S | \mathcal{O}_h^{(L)})]$. $\square$

Remark 3.3 (Finite analogue of collective exhaustion). Corollary 3.2 and its converse (Corollary 5.12, proved under fibre mixing in §5) together say: algebraic indistinguishability $\delta(L)$ vanishes if and only if the mass of indistinguishable pairs $(\mu \otimes \mu)(R_L)$ vanishes. This is the finite-sample counterpart of collective exhaustion [5]. There, CE asserts that set-theoretic vanishing must be witnessed at some finite level of the refinement order. Here, the analogous principle is that observational indistinguishability cannot persist globally without manifesting as a non-vanishing mass of indistinguishable pairs at finite lag L . The fibre mixing condition

plays an analogous structural role to CE: it is a sufficient condition under which the global and pairwise witnesses are comparable. Whether fibre mixing is itself derivable from dynamical conditions — or is genuinely irreducible, as CE is proved to be in [5] — remains open; see the open problems in Section 8.

Lemma 3.4 (Bias bound). *For every $f \in L^\infty(\mu)$,*

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S \triangle E) \leq \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2 -best $\mathfrak{m}$ -measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \leq \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 = \mu(S \triangle E) \leq \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \leq M$, the layer cake representation gives $f = \int_{-M}^M \mathbb{1}_{S_t} dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2 ,

$$f - \mathbb{E}[f \mid \mathfrak{m}] = \int_{-M}^M (\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]) dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} \leq \int_{-M}^M \|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\|_{L^2} dt \leq 2M \cdot \delta(\mathfrak{m})^{1/2}. \quad \square$$

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ -algebra at lag L , where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 3.5 (Three-term decomposition). *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{\text{bias}} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{\text{approx.}} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{\text{statistical}}, \quad (3)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (??) and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 3.4. $\square$

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n .

Lemma 3.6 (Observable reverse lower bound). *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \rightarrow \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (??) at lag L and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ is the empirical measure.

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2 -projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n , $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C'n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

4 Concentration of the Empirical Witness

Throughout this section: (T, h) satisfies (G1)–(G2), X is a compact smooth d -manifold, μ has smooth positive density, and $d > 2s$.

4.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \leq 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 4.1 (Uniform boundedness). *Define*

$$M_{D,L} := \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \text{essinf}_X \rho > 0$ is the infimum of the density of μ . Then for every $p \in \mathcal{P}_D^{(L)}$:

$$\|p\|_{L^\infty} \leq M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to the unit box by setting $y_i = x_i/a$; polynomials of degree $\leq D$ in $x \in [-a, a]^{L+1}$ correspond to polynomials $\tilde{p}$ of degree $\leq D$ in $y \in [-1, 1]^{L+1}$. Expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \leq D}$, where $P_\alpha(y) = \prod_{i=1}^{L+1} P_{\alpha_i}(y_i)$ and $\|P_k\|_{L^\infty[-1,1]} = 1$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives

$$|\tilde{p}(y)| \leq \sum_{|\alpha| \leq D} |c_\alpha| \leq N(D, L+1)^{1/2} \left(\sum_{|\alpha| \leq D} |c_\alpha|^2 \right)^{1/2},$$

where $N(D, L+1) = \binom{D+L+1}{L+1}$ counts the basis elements. From the L^2 orthogonality $\|P_\alpha\|_{L^2[-1,1]}^2 = \prod_i (2\alpha_i + 1)^{-1}$, the coefficient norm satisfies

$$\sum_{|\alpha| \leq D} |c_\alpha|^2 \leq (2D+1) \|\tilde{p}\|_{L^2}^2,$$

since the maximum of $\prod_i (2\alpha_i + 1)$ over $|\alpha| \leq D$ is $2D+1$, achieved at $\alpha = (D, 0, \dots, 0)$ — the constraint $|\alpha| \leq D$ forces all weight onto one coordinate, giving an L -independent maximum. Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\text{vol})}$ and $\|p\|_{L^2(\text{vol})} \leq \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. The subsequent arguments require $\|h\|_{L^\infty} \geq \frac{1}{2}$, equivalently $2a \geq 1$, so that $M_{D,L}$ grows at most polynomially in L . $\square$

4.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} = \left\{ x \mapsto (p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x))^2 : p \in \mathcal{P}_D^{(L)}, \|p\|_{L^2(\mu)} \leq 1 \right\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 4.5). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 4.2 (Concentration, $\star$). *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \geq 1$:*

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 \cdot D^d}\right). \quad (\star)$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C' n^{-2s/(2s+d)}$:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > \varepsilon_n/2) \leq 2 \exp\left(-c' n^{(d-2s)/(2s+d)}\right) \rightarrow 0.$$

Proof. Step 1 (Boundedness). By Lemma 4.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$ satisfies $\|p\|_{L^\infty} \leq M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \leq 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x'_i changes $\hat{\delta}(L, n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality:

$$P(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t) \leq 2 \exp\left(-\frac{2n^2 t^2}{n \cdot (B_{L,D}^2/n)^2}\right) = 2 \exp\left(-\frac{n t^2}{2B_{L,D}^4/n}\right).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [2]:

$$\mathbb{E}[\hat{\delta}(L, n)] - \delta(L) \leq 2 \mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma, since $f = (p - \cdot)^2$ is $2B_{L,D}$ -Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq 2B_{L,D} \mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d -dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [3]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \leq C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq C M_{D,L} \sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P(|\hat{\delta}(L, n) - \delta(L)| > t) \leq 2 \exp\left(-\frac{c n t^2}{M_{D,L}^4 D^d}\right).$$

Setting $t = \varepsilon_n/2 = C' n^{-2s/(2s+d)}/2$, the exponent becomes

$$c n \varepsilon_n^2 / (M_{D,L}^4 D^d) \propto n^{1-4s/(2s+d)} = n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. $\square$

Remark 4.3 (The $d > 2s$ hypothesis). When $d \leq 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [1] is technically available but left to future work; we take $d > 2s$ as a standing hypothesis.

4.3 Source 1: the LLN gap

Lemma 4.4 (LLN gap). *With notation as above:*

$$\sup_{p \in \mathcal{P}_D^{(L)}} \left| \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2 - \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \right| \leq C \sqrt{D^d/n}$$

almost surely as $n \rightarrow \infty$, where C depends on $(d, D, M_{D,L})$.

Proof. The class $\{(p - \mathbb{E}[p \mid \Phi_h^{(L)}])^2 : p \in \mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [3] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. $\square$

4.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 4.5 (Nadaraya–Watson estimator). For bandwidth $h_n > 0$ and kernel $K : \mathbb{R}^{L+1} \rightarrow \mathbb{R}_{\geq 0}$ (compactly supported, symmetric, $\int K = 1$):

$$\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}(x)] := \frac{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right) p(x_j)}{\sum_{j=1}^n K\left(\frac{\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x_j)}{h_n}\right)}.$$

Proposition 4.6 (Source 2 estimator bias). *Assume:*

- (i) $T, h \in C^r$ with $r \geq 2$, μ smooth positive density ρ with $\rho_{\min} > 0$.
- (ii) $\Phi_h^{(L)}$ has full rank μ -a.e. (follows from (G1)–(G2) and Lemma 5.2).
- (iii) T is exponentially mixing with rate $\lambda > 0$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$.
- (iv) The NW bandwidth satisfies $h_n \sim n^{-1/(2\beta + d_{\text{eff}}(L))}$ with $d_{\text{eff}}(L) = \min(L+1, d)$ and $\beta \geq 1$.

Then there exists $C_2(D, L) > 0$ such that for all $p \in \mathcal{P}_D^{(L)}$:

$$\mathbb{E} \|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq C_2(D, L) \cdot n^{-2\beta/(2\beta + d_{\text{eff}}(L))},$$

uniformly in $p \in \mathcal{P}_D^{(L)}$, where $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$ and A is a constant depending on $(\beta, d_{\text{eff}}(L), \rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu = \int \mu_z d((\Phi_h^{(L)})_* \mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z = \{\Phi_h^{(L)} = z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z - \mu_{z'}\|_{\text{TV}} \leq C_1 |z - z'|^\beta, \quad \beta = 1 \text{ (Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r}, \|h\|_{C^r}, \rho_{\min})$. (The optimal exponent $\beta = r - 1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta = 1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 4.1: for $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$,

$$|\mathbb{E}[p \mid \Phi_h^{(L)} = z] - \mathbb{E}[p \mid \Phi_h^{(L)} = z']| = \left| \int p d\mu_z - \int p d\mu_{z'} \right| \leq M_{D,L} \cdot C_1 \cdot |z - z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p \mid \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L}C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [3] for a Hölder(β) regression function on a $d_{\text{eff}}(L)$ -dimensional domain gives:

$$\mathbb{E} \|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \leq \Lambda_{D,L}^2 h_n^{2\beta} + \frac{C_K M_{D,L}^2}{n h_n^{d_{\text{eff}}(L)} f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_*\mu$. The $M_{D,L}^2$ factor appears in both terms. Balancing: $\Lambda_{D,L}^2 h_n^{2\beta} = C_K M_{D,L}^2 / (n h_n^{d_{\text{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta+d_{\text{eff}}(L))}$ where A depends only on $(\beta, d_{\text{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L , or p . The $M_{D,L}^2$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^* , both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\text{eff}}(L))}$ with $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$. Since the bound is p -free, it holds uniformly over $\mathcal{P}_D^{(L)}$. $\square$

Theorem 4.7 (Full convergence $\hat{\delta}(L, n) \rightarrow \delta(L)$). *Under the hypotheses of Proposition 4.6 and with $d > 2s$:*

$$\mathbb{E} |\hat{\delta}(L, n) - \delta(L)| \leq C \sqrt{D^d/n} + C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$$

and $P(|\hat{\delta}(L, n) - \delta(L)| > \varepsilon_n) \rightarrow 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L, n) - \delta(L)| \leq \underbrace{|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]|}_{\text{concentration, } (\star)} + \underbrace{|\mathbb{E}[\hat{\delta}(L, n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C \sqrt{D^d/n}$ (Lemma 4.4); Source 2 is bounded by $C_2(D, L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\text{eff}}(L))}$ (Proposition 4.6). The concentration term is bounded by $(\star)$ (Proposition 4.2). Combining at $t = \varepsilon_n/2$ gives the probability statement. $\square$

5 The Algebra Theorem

This section assembles the results of Sections 3 and 4 into the Algebra Theorem: the data-driven stopping rule $\hat{L}^*$ achieves the minimax-optimal rate $n^{-s/(2s+d)}$ without knowing the oracle lag $L^*(n)$ or the mixing rate of T .

5.1 Effective dimension and the piecewise rate

Definition 5.1 (Effective dimension). The *effective dimension at lag L* is

$$d_{\text{eff}}(L) := \text{rank}(D\Phi_h^{(L)}|_x) \quad \text{at a generic point } x \in X.$$

(The rank is constant on an open dense set by the rank theorem, so this is well-defined μ -a.e.)

Lemma 5.2 (d_{eff} step function). *Under (G1)–(G2) with X a compact smooth d -manifold and μ smooth positive density:*

$$d_{\text{eff}}(L) = \min(L + 1, d) \quad \text{for all } L \geq 0.$$

Proof. The upper bound $d_{\text{eff}}(L) \leq \min(L + 1, d)$ holds because $D\Phi_h^{(L)}|_x$ is a linear map from $T_x X$ (dimension d) to $\mathbb{R}^{L+1}$.

For the lower bound we use the Sard–Smale theorem (Smale [6] framework) applied to two failure maps.

Case $L + 1 \leq d$. Rank $< L + 1$ at x means the cotangent vectors $\omega_j(x) := (DT^j|_x)^*(dh|_{T^j x})$ are linearly dependent. Define the failure map

$$\Psi : C^2(X) \times X \times \mathbb{P}^L \rightarrow T^*X, \quad (h, x, [\lambda]) \mapsto d\left(\sum_j \lambda_j h \circ T^j\right)|_x.$$

By (G1), the orbit points $T^j x$ are distinct μ -a.e., so bump functions at $T^j x$ can prescribe any cotangent vector at x independently. Hence $D_h \Psi$ is surjective onto $T_x^* X$ (dimension d), and $\Psi^{-1}(0)$ is a Banach submanifold of codimension d . The projection to $C^2(X)$ is Fredholm of index $d + L - d = L$, so by Sard–Smale the regular values form a residual set $R \subset C^2(X)$. For $h \in R$, the failure locus in X has dimension $\leq L < d$, hence μ -measure zero.

By (G2), $dh \neq 0$ μ -a.e., which forces every critical point of $F_\lambda = \sum_j \lambda_j h \circ T^j$ to lie in the codimension- d submanifold $S_\lambda = \{dh_x \in \text{span}\{\omega_1(x), \dots, \omega_L(x)\}\}$ (a finite set for each $[\lambda]$, since X is d -dimensional). Hence h is a regular value of π , i.e., $h \in R$.

Case $L + 1 > d$. Immersion failure requires $\exists v \neq 0$ with $D\Phi_h^{(L)}|_x \cdot v = 0$. Define

$$\Xi : C^2(X) \times (TX \setminus 0) \rightarrow \mathbb{R}^{L+1}, \quad (h, (x, v)) \mapsto (dh_{T^j x}(DT^j|_x \cdot v))_{j=0}^L.$$

By (G1), $D_h \Xi$ is surjective onto $\mathbb{R}^{L+1}$, so $\Xi^{-1}(0)$ has codimension $L + 1$ and the projection to $C^2(X)$ is Fredholm of index $2d - 1 - (L + 1) = 2d - L - 2$. For $L \geq d$: $2d - L - 2 \leq d - 2 < d - 1$, so the failure locus in X has dimension $< d - 1$ and μ -measure zero. Condition (G2) places h in the regular-value set by the same argument as Case 1. $\square$

Proposition 5.3 (Piecewise pre-Takens rate). *Assume (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{H\"older}(s) \cap L^\infty(\mu)$, $d > 2s$, and $T \in C^r$ with $r \geq 2$. Let $\hat{f}_n^{(L)}$ be the empirical risk minimiser over $\mathcal{A}_h^{(L)}$ with $D = D^*(n, L)$ chosen to balance approximation and statistical terms. Then for each $L \geq 0$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d_{\text{eff}}(L))}$$

with probability tending to 1, where $d_{\text{eff}}(L) = \min(L + 1, d)$. In the Takens regime ($L \geq d - 1$): $d_{\text{eff}}(L) = d$ and the rate locks at $n^{-s/(2s+d)}$.

Proof. Apply the three-term decomposition (Lemma 3.5). The bias term is bounded by $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}$ (Lemma 3.4). By Lemma 5.2, $\Phi_h^{(L)}(X)$ is a $d_{\text{eff}}(L)$ -dimensional smooth submanifold. On this submanifold, $\mathbb{E}[f | \mathcal{O}_h^{(L)}](x) = g^*(\Phi_h^{(L)}(x))$ for some $g^* \in \text{H\"older}(s)$; standard polynomial approximation on a $d_{\text{eff}}(L)$ -dimensional smooth domain gives approximation error $\leq C_{\text{approx}} \cdot D^{-s/d_{\text{eff}}(L)}$. The statistical term is $\leq C_{\text{stat}}(D^{d_{\text{eff}}(L)}/n)^{1/2}$ by Proposition 4.2 with d replaced by $d_{\text{eff}}(L)$. Balancing: $D^* \sim n^{d_{\text{eff}}(L)/(2s+d_{\text{eff}}(L))}$ and both terms are $O(n^{-s/(2s+d_{\text{eff}}(L))})$. $\square$

5.2 The oracle lag and the elbow stopping rule

Definition 5.4 (Oracle lag). The *oracle lag* is

$$L^*(n) := \min\{L \geq 0 : \delta(L) \leq C_0 \cdot n^{-2s/(2s+d)}\},$$

where $C_0 > 0$ is a constant depending on $\|f\|_{L^\infty}$. At $L = L^*(n)$, the bias term $2\|f\|_{L^\infty} \cdot \delta(L^*)^{1/2}$ is $O(n^{-s/(2s+d)})$, matching the statistical floor.

The oracle lag $L^*(n)$ requires knowledge of the mixing rate, which is unobservable. The empirical witness $\hat{\delta}(L, n)$ sidesteps this: it measures reconstruction quality directly from data.

Definition 5.5 (Elbow stopping rule). Fix a tolerance sequence $\tau_n > 0$. The *elbow stopping rule* is

$$\hat{L}^* := \min\{L \geq 0 : \hat{\delta}(L + 1, n) - \hat{\delta}(L, n) > -\tau_n\}.$$

Theorem 5.6 (Elbow location). *Assume the hypotheses of Proposition 5.3 and exponential mixing $\delta(L) \leq Ae^{-\lambda L}$ with resolvability constant $C_\lambda := (1 - e^{-\lambda})Ae^\lambda/C' > 4$. Set $\tau_n = \frac{C_\lambda}{2} \cdot C' \cdot n^{-2s/(2s+d)}$. Then*

$$P(\hat{L}^* = L^*(n)) \rightarrow 1 \quad \text{exponentially in } n.$$

Proof. We show the stopping criterion fires at $L = L^*(n)$ (no overshoot) and not before (no early stop).

No overshoot: $\hat{L}^* \leq L^*(n)$. At $L = L^*(n)$, the population increment satisfies $|\delta(L^*(n) + 1) - \delta(L^*(n))| \leq \delta(L^*(n)) \leq C_0 n^{-2s/(2s+d)}$ (non-increasing δ). By Theorem 4.7, $|\hat{\delta}(L, n) - \delta(L)| \leq \varepsilon_n = C' n^{-2s/(2s+d)}$ whp. The empirical increment satisfies $\hat{\delta}(L^* + 1, n) - \hat{\delta}(L^*, n) \geq -C_0 n^{-2s/(2s+d)} - 2\varepsilon_n > -\tau_n$ whp, so the criterion fires.

No early stop: $\hat{L}^* \geq L^*(n)$. For $L < L^*(n)$, exponential mixing gives population decrement $\delta(L) - \delta(L + 1) \geq \Delta(L) \geq C_\lambda \cdot C' n^{-2s/(2s+d)}$. Tracking via Theorem 4.7:

$$\hat{\delta}(L, n) - \hat{\delta}(L + 1, n) \geq C_\lambda C' n^{-2s/(2s+d)} - 2\varepsilon_n = (C_\lambda - 2)C' n^{-2s/(2s+d)}.$$

Since $C_\lambda > 4$: $(C_\lambda - 2)C' > \tau_n \cdot 2/C_\lambda \cdot C_\lambda = 2\tau_n/C_\lambda \cdot C_\lambda$, and specifically $(C_\lambda - 2)C' n^{-2s/(2s+d)} \geq \tau_n$ when $C_\lambda - 2 \geq C_\lambda/2$, i.e., $C_\lambda \geq 4$. So the stopping criterion does *not* fire for $L < L^*(n)$, and $\hat{L}^* \geq L^*(n)$.

Remark 5.7 (Scope of the elbow theorem). The resolvability condition $C_\lambda > 4$ is not an artefact: when $C_\lambda \leq 4$ the elbow drop at $L^*(n)$ merges with the noise floor ε_n and the two are indistinguishable from data. For polynomial mixing, $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \rightarrow 0$ relative to ε_n , so the elbow drops vanish asymptotically and the theorem fails. A separate argument (e.g. integrated elbow) is needed for that case and is not proved here. □

Theorem 5.8 (Algebra Theorem). *Assume:*

- (i) *Hypotheses of Proposition 5.3: (G1)–(G2), X compact smooth d -manifold, μ smooth positive density, $f \in \text{H\"older}(s) \cap L^\infty(\mu)$, $d > 2s$, $T \in C^r$, $r \geq 2$.*
- (ii) *Exponential mixing: $\delta(L) \leq Ae^{-\lambda L}$ for some $A, \lambda > 0$.*
- (iii) *Resolvability: $C_\lambda > 4$.*
- (iv) *Exponential mixing for the estimator bias (Proposition 4.6) and $\|h\|_{L^\infty} \geq \frac{1}{2}$.*

Let $\hat{L}^$ be the elbow stopping rule (Definition 5.5) with $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$. Then*

$$\|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} = O_p(n^{-s/(2s+d)}).$$

The rate $n^{-s/(2s+d)}$ is minimax-optimal for $\text{H\"older}(s)$ functions on a d -dimensional domain. No oracle input (mixing rate, $L^(n)$, Sobolev index s) is required.*

Proof. On the event $\{\hat{L}^* = L^*(n)\}$, which has probability tending to 1 by Theorem 5.6:

$$\begin{aligned} \|f - \hat{f}_n^{(\hat{L}^*)}\|_{L^2(\mu)} &= \|f - \hat{f}_n^{(L^*(n))}\|_{L^2(\mu)} \\ &\leq 2\|f\|_{L^\infty} \cdot \delta(L^*(n))^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &\leq 2\|f\|_{L^\infty} \cdot (C_0 n^{-2s/(2s+d)})^{1/2} + C(f, d, s) \cdot n^{-s/(2s+d)} \\ &= O(n^{-s/(2s+d)}). \end{aligned} \quad \square$$

5.3 Entropy characterisation of reconstruction

Definition 5.9 (Fibre mixing). Fix $\varepsilon > 0$. An event $S^* \in \mathcal{B}$ is (c, ε) -mixing across fibres if there exists $c > 0$ such that

$$\mu_z(S^*) \mu_z((S^*)^c) \geq c \cdot p_z \quad \text{for } \nu_L\text{-a.e. } \varepsilon\text{-large fibre } z.$$

Ergodicity of T is expected to force this condition (see Section 8), but the general derivability question remains open; see the open problems below. This condition is independent of (US) and is not a standing assumption here; it is used only in Corollary 5.12 below.

Lemma 5.10 (Positive-fraction balance). Let $S^* \in \mathcal{B}$ be an ε -near-maximizer of $\delta(L)$, i.e. $\mathbb{E}[\text{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \geq \delta(L) - \varepsilon$. Define the large-fibre contribution

$$\kappa := \int_{\{p_z \geq \varepsilon\}} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L(z).$$

For any $\eta \in (0, \frac{1}{2})$ with $\kappa > \eta$, the set

$$G(\eta) := \{z : p_z \geq \varepsilon, \mu_z(S^*) \mu_z((S^*)^c) \geq \eta\}$$

of ε -large balanced fibres satisfies

$$\nu_L(G(\eta)) \geq 4(\kappa - \eta).$$

No dynamical hypothesis is required.

Proof. Partition the ε -large fibres into monochromatic fibres $\mathcal{M} = \{z : p_z \geq \varepsilon, \mu_z(S^*) \mu_z((S^*)^c) < \eta\}$ and balanced fibres $G(\eta)$. On $\mathcal{M}$: $\mu_z(S^*) \mu_z((S^*)^c) < \eta$, so

$$\int_{\mathcal{M}} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \leq \eta \int_{\mathcal{M}} p_z d\nu_L \leq \eta.$$

Hence $\int_{G(\eta)} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \geq \kappa - \eta$. On $G(\eta)$: $\mu_z(S^*) \mu_z((S^*)^c) \leq \frac{1}{4}$ pointwise, so

$$\kappa - \eta \leq \int_{G(\eta)} p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L \leq \frac{1}{4} \int_{G(\eta)} p_z d\nu_L \leq \frac{1}{4} \nu_L(G(\eta)). \quad \square$$

Lemma 5.11 (Fibre mixing bound). Let S^* be a $(\delta(L) - \varepsilon)$ -near-maximizer of $\mathbb{E}[\text{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})]$ that is (c, ε) -mixing across fibres (Definition 5.9). Then

$$(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L) + C(c) \varepsilon, \quad C(c) := \frac{1+c}{c}.$$

Proof. Write $(\mu \otimes \mu)(R_L) = \int p_z^2 d\nu_L(z)$ and split into ε -large and small-fibre parts.

Large fibres ($p_z \geq \varepsilon$). The (c, ε) -mixing condition gives $\mu_z(S^*) \mu_z((S^*)^c) \geq c p_z$ on ε -large fibres, so:

$$c \int_{\{p_z \geq \varepsilon\}} p_z^2 d\nu_L(z) \leq \int p_z \mu_z(S^*) \mu_z((S^*)^c) d\nu_L(z) = \mathbb{E}[\text{Var}(\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)})] \leq \delta(L).$$

Hence $\int_{\{p_z \geq \varepsilon\}} p_z^2 d\nu_L(z) \leq \frac{1}{c} \delta(L)$.

Small fibres ($p_z < \varepsilon$). Each such fibre satisfies $p_z^2 \leq \varepsilon p_z$, so

$$\int_{\{p_z < \varepsilon\}} p_z^2 d\nu_L(z) \leq \varepsilon \int p_z d\nu_L(z) = \varepsilon.$$

Combining. Since $C(c) = \frac{1+c}{c} \geq 1$ we have $\varepsilon \leq C(c) \varepsilon$, and therefore

$$(\mu \otimes \mu)(R_L) \leq \frac{1}{c} \delta(L) + C(c) \varepsilon. \quad \square$$

The collision probability (??) defines a natural information-theoretic quantity:

$$H_2(\nu_L) := -\log((\mu \otimes \mu)(R_L)) = -\log \int p_z^2 d\nu_L(z). \quad (4)$$

This is the *Rényi-2 (collision) entropy* of the delay-vector distribution.

Corollary 5.12 (Entropy characterisation of reconstruction). *Under the fibre mixing condition of Definition 5.9:*

$$\delta(L) \rightarrow 0 \iff H_2(\nu_L) \rightarrow +\infty.$$

Equivalently, reconstruction occurs if and only if the probability that two independently drawn delay vectors coincide vanishes as $L \rightarrow \infty$. Thus reconstruction is equivalent to the vanishing of collision probability $(\mu \otimes \mu)(R_L)$ introduced in Section 2.4.

Proof. Corollary 3.2 gives $\delta(L) \leq \frac{1}{2}(\mu \otimes \mu)(R_L)$. For the reverse, Lemma 5.11 with a $\delta(L)$ -near-maximizer S^* satisfying the (c, ε) -mixing condition gives $(\mu \otimes \mu)(R_L) \leq \frac{1}{c}\delta(L) + C(c)\varepsilon$, so $\delta(L) \rightarrow 0 \iff (\mu \otimes \mu)(R_L) \rightarrow 0$. Since $(\mu \otimes \mu)(R_L) = e^{-H_2(\nu_L)}$, this is equivalent to $H_2(\nu_L) \rightarrow +\infty$. $\square$

Remark 5.13 (Balance mechanism). Lemma 5.10 guarantees that a positive ν_L -fraction of ε -large fibres are η -balanced without any dynamical hypothesis. The fibre mixing condition is the upgrade from positive-fraction to ν_L -a.e. balance; whether ergodicity supplies it is the main open question (Section 8).

Remark 5.14 (Empirical entropy witness). The empirical analogue of $H_2(\nu_L)$ is

$$\hat{H}_2(\nu_L^{(n)}) := -\log \left(\frac{1}{n^2} \sum_{i,j=1}^n \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)] \right),$$

the negative log of the empirical collision probability. This is the third computable witness for reconstruction. It is *computationally simpler* than $\hat{\delta}(L, n)$: it requires no optimisation over sets, only histogram counts over delay vectors. Under fibre mixing, Corollary 5.12 shows that $\hat{L}^*$ could equivalently be defined as the first L at which $\hat{H}_2(\nu_L^{(n)})$ exceeds a threshold calibrated to n .

6 The Dynamics Theorem

6.1 Lipschitz continuity of the delay map

Lemma 6.1 (Upper Lipschitz bound). *Assume $T \in \text{Diff}^r(X)$, $h \in C^r(X)$, $r \geq 1$. Then $\Phi_h^{(L)}$ is Lipschitz:*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \leq C_L \cdot d_X(x, x') \quad \forall x, x' \in X,$$

where $C_L := \|Dh\|_{L^\infty} \cdot (\sum_{j=0}^L \|DT^j\|_{L^\infty}^2)^{1/2}$.

Proof. Component j of $\Phi_h^{(L)}$ is $h \circ T^j$. By the chain rule and compactness of X : $|h(T^j x) - h(T^j x')| \leq \|Dh\|_{L^\infty} \cdot \|DT^j\|_{L^\infty} \cdot d_X(x, x')$. Taking Euclidean norm over $j = 0, \dots, L$ gives the result. $\square$

Lemma 6.2 (Local lower Lipschitz bound). *Under (G1)–(G2) with $L \geq d-1$ (Takens regime), $T \in C^r$, $r \geq 2$: for μ -a.e. $x \in X$, there exists a neighbourhood U_x and a constant $c_x = \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ such that*

$$|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c_x \cdot d_X(x, x') \quad \forall x' \in U_x.$$

Proof. By Lemma 5.2, $D\Phi_h^{(L)}|_x$ has full rank d for μ -a.e. x when $L \geq d-1$, so $\sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$ μ -a.e. The lower bound on U_x follows from the inverse function theorem applied to the immersion $\Phi_h^{(L)}$ at x . $\square$

Remark 6.3 (Gap between local and global lower bound). Lemma 6.2 gives a *local* lower bound $c_x > 0$ varying over X . (G1)–(G2) guarantee $\sigma_{\min} > 0$ μ -a.e., but not on all of X . Condition (US) imposes $c := \inf_X \sigma_{\min}(D\Phi_h^{(L)}|_x) > 0$, which is strictly stronger. The gap is genuine: there exist pairs (T, h) satisfying (G1)–(G2) but violating (US) on a μ -null set.

6.2 Uniform separation

The one-step predictive kernel for deterministic T is the Dirac family $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$. The TV separation pseudometric is therefore

$$d_L(x, x') := \|\Pi_h^{(L)}(x, \cdot) - \Pi_h^{(L)}(x', \cdot)\|_{\text{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')] \in \{0, 1\}.$$

Separation is binary: $d_L(x, x') = 1$ if and only if x and x' are distinguished by $\mathcal{O}_h^{(L)}$.

Theorem 6.4 (Uniform separation under (US)). *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, $L \geq d-1$. Then $\Phi_h^{(L)}$ is bi-Lipschitz on X :*

$$c \cdot d_X(x, x') \leq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \leq C_L \cdot d_X(x, x') \quad \forall x, x' \in X.$$

For any estimator $\hat{\Gamma}_h^{(n)}$ with uniform error $|\hat{\Gamma}_h^{(n)}(x) - \Phi_h^{(L)}(x)| \leq \varepsilon_n$ uniformly in x , the empirical separation $\hat{d}_L(x, x') := |\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')|$ satisfies

$$\hat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n.$$

In particular, if $d_L(x, x') = 1$ (i.e. $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$), then $\hat{d}_L(x, x') > 0$ for all n large enough that $\varepsilon_n < c \cdot d_X(x, x')/2$.

Proof. Upper bound: Lemma 6.1. Lower bound: (US) directly gives $|\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| \geq c \cdot d_X(x, x')$ for all $x, x' \in X$. For the empirical bound: by the triangle inequality, $|\hat{\Gamma}_h^{(n)}(x) - \hat{\Gamma}_h^{(n)}(x')| \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')| - 2\varepsilon_n \geq c \cdot d_X(x, x') - 2\varepsilon_n$. If $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ then by bi-Lipschitz $d_X(x, x') \geq |\Phi_h^{(L)}(x) - \Phi_h^{(L)}(x')|/C_L > 0$, so the right side is eventually positive. $\square$

Theorem 6.5 (Dynamics Theorem). *Assume (G1)–(G2), (US), $T \in C^r$, $r \geq 2$, μ smooth positive density on compact d -manifold X , $L \geq d-1$. Let $\hat{\Gamma}_h^{(n)}$ be the empirical delay map estimator with uniform error $\varepsilon_n = C'' n^{-\beta/(2\beta+d)}$ (Proposition 4.6). Then:*

- (a) (Convergence) $\|\hat{\Gamma}_h^{(n)} - \Phi_h^{(L)}\|_{L^\infty} \rightarrow 0$ at rate $n^{-\beta/(2\beta+d)}$.
- (b) (Separation) For $\mu \otimes \mu$ -a.e. (x, x') with $x \neq x'$: $\hat{d}_L(x, x') > 0$ for all large n .
- (c) (Scope) Condition (US) is not a consequence of (G1)–(G2). Without (US), the kernel witness $\hat{d}_L$ can vanish on a positive-measure set of pairs even when $d_L = 1$ and reconstruction holds.

Proof. (a) Proposition 4.6 with $d_{\text{eff}}(L) = d$ gives $\varepsilon_n \rightarrow 0$ at rate $n^{-\beta/(2\beta+d)}$.

(b) Since μ is non-atomic (smooth positive density), $\mu \otimes \mu$ almost every pair satisfies $d_X(x, x') > 0$. For such pairs $d_L(x, x') = 1$ (reconstruction and injectivity of $\Phi_h^{(L)}$), and by Theorem 6.4, $\hat{d}_L(x, x') \geq c \cdot d_X(x, x') - 2\varepsilon_n$. For fixed $x \neq x'$, $c \cdot d_X(x, x') > 0$ and $\varepsilon_n \rightarrow 0$, so the right side is eventually positive.

(c) The proof of Theorem 6.4 uses (US) in the lower bound. Without it, there exist points x where $\sigma_{\min}(D\Phi_h^{(L)}|_x) \rightarrow 0$ along sequences $x'_n \rightarrow x$, and $|\Phi_h^{(L)}(x'_n) - \Phi_h^{(L)}(x)|/d_X(x, x'_n) \rightarrow 0$. Any estimator inherits this degeneracy. $\square$

7 The Conjunction Theorem

This section formalises the complete reconstruction diagnostic and proves that both witnesses certify the same event when reconstruction and (US) hold.

7.1 The honest bridge

The delay vector $\Phi_h^{(L)}(x) = (h(x), \dots, h(T^L x))$ is a trajectory of the Markov chain on h -space with one-step kernel Π_h . The σ -algebra $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ is exactly the information generated by L steps of this Markov chain starting from x . This gives an algebraic identity:

$$\sigma(\Phi_h^{(L)}) = \mathcal{O}_h^{(L)},$$

which is not an approximation but a definition. Consequently, the σ -algebra separation event and the metric separation event are identical:

Lemma 7.1 (Algebra separation equals metric separation). *For deterministic T and $\mu \otimes \mu$ -a.e. (x, x') :*

$$d_L(x, x') = 1 \iff x \text{ and } x' \text{ are separated by } \mathcal{O}_h^{(L)}.$$

These are the same event, not merely correlated.

Proof. For deterministic T : $\Pi_h^{(L)}(x, \cdot) = \delta_{\Phi_h^{(L)}(x)}$, so $d_L(x, x') = \|\delta_{\Phi_h^{(L)}(x)} - \delta_{\Phi_h^{(L)}(x')}\|_{\text{TV}} = \mathbb{1}[\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')]$. And x is separated from x' by $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ if and only if $\exists E \in \mathcal{O}_h^{(L)}$ with $\mathbb{1}_E(x) \neq \mathbb{1}_E(x')$, i.e., if and only if $\Phi_h^{(L)}(x) \neq \Phi_h^{(L)}(x')$ (since $\mathcal{O}_h^{(L)}$ is generated by the level sets of $\Phi_h^{(L)}$). The two conditions are identical. $\square$

Under reconstruction ($\delta(L) = 0$, equivalently $\Phi_h^{(L)}$ injective μ -a.e.) and (US), injectivity holds everywhere, so $d_L(x, x') = 1$ for $\mu \otimes \mu$ -a.e. $x \neq x'$.

7.2 The joint diagnostic

Definition 7.2 (Joint reconstruction diagnostic). Fix tolerances $\tau_n, \eta_n > 0$. Define

$$\text{Reconstruct}(L, n) := \{\hat{\delta}(L, n) \leq \tau_n\} \wedge \{\hat{d}_L(x, x') > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}\}.$$

Theorem 7.3 (Conjunction Theorem). *Assume the hypotheses of Theorem 5.8 (for the algebra side) and Theorem 6.5 (for the dynamics side). Set $\tau_n = \frac{C_\lambda}{2} C' n^{-2s/(2s+d)}$ and $\eta_n = \sqrt{\varepsilon_n}$ where $\varepsilon_n = C' n^{-2s/(2s+d)}$. Then:*

(a) (Conjunction fires under reconstruction + (US)). *Under $\delta(L) = 0$ for $L \geq L^*(n)$ and (US):*

$$P(\text{Reconstruct}(\hat{L}^*, n)) \rightarrow 1 \text{ exponentially in } n.$$

(b) (Algebra witness fails when reconstruction fails). *If $\delta(L) > 0$ for all L : $P(\hat{\delta}(L, n) \leq \tau_n) \rightarrow 0$ for every fixed L , and $\hat{L}^* \rightarrow \infty$.*

(c) (Kernel witness fails when (US) fails). *If $\sigma_{\min}(D\Phi_h^{(L)}|_x) \rightarrow 0$ on a positive-measure set S : $P(\hat{d}_L > \eta_n \text{ for } \mu \otimes \mu\text{-most pairs}) \rightarrow 0$ even when $\hat{\delta}(L, n) \rightarrow 0$.*

In particular: the conjunction is not achievable when either hypothesis fails.

Proof. Part (a). By Theorem 5.6, $P(\hat{L}^* = L^*(n)) \rightarrow 1$ exponentially. On $\{\hat{L}^* = L^*(n)\}$: (i) $\hat{\delta}(\hat{L}^*, n) \leq \tau_n$ whp by Theorem 4.7 and the definition of $L^*(n)$ ($\delta(L^*(n)) = 0$ under reconstruction). (ii) By Theorem 6.5(b), $\hat{d}_{\hat{L}^*}(x, x') > 0$ for $\mu \otimes \mu$ -a.e. (x, x') with $x \neq x'$, eventually. The set $\{d_X(x, x') \leq (2\varepsilon_n + \eta_n)/c\}$ has $\mu \otimes \mu$ -measure $\rightarrow 0$ as $\eta_n/c \rightarrow 0$, so $\hat{d}_{\hat{L}^*} > \eta_n$ for $\mu \otimes \mu$ -most pairs.

Part (b). If $\delta(L) > 0$ for all L , then by Theorem 4.7, $\hat{\delta}(L, n) \geq \delta(L) - \varepsilon_n \geq \delta(L)/2 > \tau_n$ with high probability for all fixed L , so the first condition of Reconstruct never fires.

Part (c). If $S = \{x : \sigma_{\min}(D\Phi_h^{(L)}|_x) = 0\}$ has $\mu(S) > 0$, then for $x \in S$ there exist sequences $x'_k \rightarrow x$ with $|\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)|/d_X(x'_k, x) \rightarrow 0$. Any consistent estimator satisfies $\hat{d}_L(x, x'_k) \leq |\Phi_h^{(L)}(x'_k) - \Phi_h^{(L)}(x)| + 2\varepsilon_n \rightarrow 0$. The set of pairs where $\hat{d}_L \leq \eta_n$ thus has positive $\mu \otimes \mu$ -measure, and the kernel condition fails. $\square$

8 Discussion

Polynomial mixing. The Algebra Theorem (Theorem 5.8) and the Elbow Location Theorem (Theorem 5.6) require exponential mixing with resolvability constant $C_\lambda > 4$. Under polynomial mixing $\delta(L) \lesssim L^{-\alpha}$, the elbow drop at $L^*(n)$ satisfies $\Delta(L^*(n)) \sim \varepsilon_n/L^*(n) \rightarrow 0$ relative to ε_n : the elbow and the noise floor merge asymptotically and cannot be separated. A different stopping criterion (e.g. integrated elbow, or a bandwidth-adaptive rule) is needed.

The $d > 2s$ hypothesis. Proposition 4.2 requires $d > 2s$ for the Rademacher concentration to be useful at rate ε_n . When $d \leq 2s$, localised Rademacher complexity (Bartlett–Bousquet–Mendelson [1]) gives adaptive concentration without this condition but requires additional work.

Uniform separation from first principles. Condition (US) is a hypothesis in Theorems 6.5 and 7.3. The natural conjecture is that T Anosov with h generic implies (US) via the uniform hyperbolicity constants. This would close the gap between (G1)–(G2) and (US) in the hyperbolic category.

Stochastic T . For stochastic dynamics, $\Pi_h^{(L)}(x, \cdot)$ is not a Dirac delta. The TV separation d_L is no longer binary, and Hölder continuity of $x \mapsto \Pi_h^{(L)}(x, \cdot)$ in TV is both necessary and non-trivial. The four-link chain of Proposition 4.6 adapts, but the Conjunction Theorem’s proof structure changes substantially.

Sharp rates for the data-driven lag. Theorem 5.6 shows $P(\hat{L}^* = L^*(n)) \rightarrow 1$ but does not bound the fluctuations of $\hat{L}^*$ around $L^*(n)$ on the event $\{\hat{L}^* \neq L^*(n)\}$. A Berry–Esseen-type result for $\hat{L}^* - L^*(n)$ would sharpen the stopping rule.

Concentration of the entropy witness. The empirical collision entropy $\hat{H}_2(\nu_L^{(n)})$ is a computationally simpler stopping criterion (Remark 5.14). Asymptotic equivalence with $\hat{\delta}(L, n)$ is established under fibre mixing (Corollary 5.12), but a finite- n concentration result for $\hat{H}_2(\nu_L^{(n)})$ — comparable to Proposition 4.2 for $\hat{\delta}(L, n)$ — is not proved here. Such a result requires only McDiarmid-type bounds on the U-statistic $\frac{1}{n^2} \sum_{i,j} \mathbb{1}[\Phi_h^{(L)}(x_i) = \Phi_h^{(L)}(x_j)]$, with no new structural assumptions beyond those already made.

Derivability of fibre mixing (the main conceptual open problem). The central open question is whether the fibre mixing condition (Definition 5.9) is derivable from dynamical hypotheses, or whether it is an irreducible admissibility condition analogous to CE in [5].

Lemma 5.10 settles Step A: a positive ν_L -fraction of ε -large fibres are η -balanced, with an explicit lower bound depending only on the large-fibre contribution κ and not on any dynamical hypothesis.

The open question is Step B: what is the weakest dynamical condition that upgrades positive-fraction balance to ν_L -a.e. balance? Ergodicity is the natural candidate: in an ergodic system, a near-maximizer of $\delta(L)$ must cross fibres rather than align with them, since aligned events are $\mathcal{O}_h^{(L)}$ -measurable and thus carry zero approximation error. The informal argument is clear, but

formalizing the connection between ergodicity and the a.e. balance condition is the isolated open gap. If ergodicity suffices, the fibre mixing condition is derivable and the structural analogy with CE breaks; if it does not, the analogy is complete.

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